

M13-C3 — H6 folding-steps and nested- n ablation (search cost under `baseline_cautious`)

H6 run / analysed / not a Bucket 3 claim

1 Status

H6 run / analysed. Bounded procedural investigation of **search cost** under repository `baseline_cautious` (`cautious + nd / any / all / relto`) on fixture `m13_bucket3_binary_collider_and`. Markdown twin: `h6_folding_and_n_ablation.md`.

H6a lead: at fixed `n30_seed42`, root (a/b) trace length is approximately linearly proportional to permitted `folding_steps( $M$ )`; extra budget beyond the failed motif is wasted search time.

H6b lead: at fixed `folding_steps(2)`, root trace length is approximately linearly proportional to sample size n (topology fixed; per-band bookkeeping grows). Roots are procedural diagnostics (evaluator `no_observed_parent_deterministic_rule`); only child c is a desirable deterministic mechanism target. H0–H5 closed / unchanged. **Not a Bucket 3 claim.** H7a analysed (`h7a_relto_vs_sechk.md`); H7b analysed separately (`h7b_cautious_relto_vs_sechk.md`).

2 Research question

Under repository-baseline `cautious + nd` on the closed H0 AND-collider table, how do permitted `folding_steps( $M$ )` (H6a) and nested sample size n at fixed $M=2$ (H6b) drive search cost for roots without recoverable deterministic parent rules, and what happens to control c ?

3 Main findings

3.1 H6a — folding-steps ablation

On frozen `n30_seed42`, for roots a/b , `prolog.stdout` length grows roughly linearly with $M \in \{1, 2, 5, 10\}$ ($\sim 549 \rightarrow 1047 \rightarrow 2541 \rightarrow 5031$ lines for a ; mirrored for b). Outcome stays `completed_no_solution` at every M . Control c solves inside `tokens(1)` with identical delta and unchanged 134-line / ~ 1.3 s trace at all four budgets.

Why predictable. `folding_steps( $M$ )` does *not* mean “perform exactly M folds.” Under `nd`, learning starts at `tokens(1)`. When a full attempt under the current token budget fails, `gen.pl` may print `* Increasing folding tokens to: T1` and retry while $T_1 \leq M$. On these roots, each budget replays essentially the same failed cautious motif (~ 9 NEW assumptions / ~ 500 -line band). Higher M multiplies failed-band replays: lines $\approx C_{\text{fixed}} + M \cdot C_{\text{band}}$. There is no successful M^* through $M=10$ — raising M does not buy a new theory; it only replays failure (H4 wasted headroom).

3.2 H6b — nested sample-size ablation

At fixed `folding_steps(2)`, for roots a/b , trace length grows roughly with $n \in \{30, 60, 90\}$ ($\sim 1047 \rightarrow 1641 \rightarrow 2241$ lines for a). Search topology is fixed ($\text{NEW}=18$, $\text{KO}=32$; one token increase to 2). What grows is per-band bookkeeping (more grounded $E^\pm$ ids; larger rote/`ert` and subsumption sweeps; larger contrary batches).

Why predictable. With motif $\times$ budget structure held fixed, cost per band should scale roughly with grounding / example size $\Rightarrow$ total length roughly $\propto n$ in this limited range. Support *presence* of the four atoms is already complete at $n=30$; multiplicities and computational grounding size change. H6b is not “pure compute with every statistical property fixed.” Line ratios are near-linear; runtime slightly more superlinear. Do not claim a universal complexity law.

3.3 Cross lead

H4’s long root traces ($\sim 5k$ lines / $\sim 66\text{--}70s$ at $n30 \times \text{steps}10$) are primarily cumulative nd token-budget replays of one cautiously failed motif (H6a), with a secondary per-band cost that grows with n at fixed $M=2$ (H6b); control c stays solved and delta-stable; nothing here is root mechanism recovery or an $n \times \text{steps}$ interaction result.

4 Setup

Fixture `m13_bucket3_binary_collider_and`. Bundle: `cautious + nd / any / all / relto`. Base Prolog `configs/baseline_cautious_config.pl` (`steps10`; unchanged); ablation Prolog `configs/baseline_cautious_steps{1,2,5}_config.pl`. H6a: fixed `n30_seed42`, $M \in \{1, 2, 5, 10\}$ (`steps10` = reused H4). H6b: fixed $M=2$; nested $n30/n60/n90$ seed 42. `n30 sha256 1edae465d0f6b6b662b3c4ed4863af21c1d9b0c1cf428774731dd3f43282a8ca`; $n30 \subset n60 \subset n90$. Atom counts $(0, 0, 0)/(0, 1, 0)/(1, 0, 0)/(1, 1, 1)$: `n30 2/4/7/17`; `n60 4/10/12/34`; `n90 5/17/18/50`.

Paths: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/<config>/<sample>`
H4 reuse summary `sha256 1e4b4199fe03a5a6208967b092450f67717d4e0dc788271f771c0d18b1ff23de`.

Representative commands:

```
python -m causal.targetwise.cli run \
  --config ../baseline_cautious_steps1/n30_seed42.yaml
python -m causal.targetwise.cli run \
  --config ../baseline_cautious_steps2/n{30,60,90}_seed42.yaml
python -m causal.targetwise.cli run \
  --config ../baseline_cautious_steps5/n30_seed42.yaml
```

Engine gate (read-only): `gen.pl` increases `tokens(T)` while $T+1 \leq M$ under `folding_mode(nd)`; initialisation `tokens(1)`.

5 Verified outcome table

Recorder-verified against `metrics.json` (`aba_learning_runtime_s`) and `output/prolog.stdout`.

sample	M	t	outcome	rt(s)	lines	incs	NEW	KO
n30	1	a	no-sol	6.638	549	none	9	16
n30	1	b	no-sol	6.797	555	none	9	16
n30	1	c	solved	1.304	134	—	1	1
n30	2	a	no-sol	13.117	1047	to 2	18	32
n30	2	b	no-sol	13.487	1062	to 2	18	32
n30	2	c	solved	1.279	134	—	1	1
n30	5	a	no-sol	32.775	2541	to 2..5	45	80
n30	5	b	no-sol	32.825	2583	to 2..5	45	80
n30	5	c	solved	1.271	134	—	1	1
n30	10	a	no-sol	69.952	5031	to 2..10	90	160
n30	10	b	no-sol	66.473	5118	to 2..10	90	160
n30	10	c	solved	1.288	134	—	1	1
n60	2	a	no-sol	22.169	1641	to 2	18	32
n60	2	b	no-sol	22.617	1659	to 2	18	32
n60	2	c	solved	2.126	200	—	1	1
n90	2	a	no-sol	32.291	2241	to 2	18	32
n90	2	b	no-sol	32.894	2262	to 2	18	32
n90	2	c	solved	3.033	266	—	1	1

Integrity: H6a n30 BK/examples/data identical across steps 1/2/5/10 per target; control c delta.aba byte-identical on all six collections (sha256 f9f75b60631b2a2a...426865a):

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

6 Analysis A (H6a)

Decisive cautious KO already inside budget-1 band:

```
folding result: c_alpha_3(A) <- [b_val_0(A)]
found: alpha_2/1 ... KO, cannot introduce an assumption for [b_val_0(A)]
```

NEW=9 $\times$ M . Steps1 body after the lopt(...) dump equals steps2 band-0 for 522 lines; then steps1 prints ** No solution found!* while steps2 prints ** Increasing folding tokens to:* 2. Rough linearity for a : lines/ $M \approx 549, 524, 508, 503$ across $M=1, 2, 5, 10$.

Control c : solved every M ; 134 lines / ~ 1.3 s; no token increases.

7 Analysis B (H6b)

Fixed $M=2$. Topology fixed (NEW=18, KO=32). a lines 1047 $\rightarrow$ 1641 $\rightarrow$ 2241; runtime 13.1 $\rightarrow$ 22.2 $\rightarrow$ 32.3s. c same delta bytes; lines 134 $\rightarrow$ 200 $\rightarrow$ 266; runtime 1.28 $\rightarrow$ 2.13 $\rightarrow$ 3.03s. Support presence complete at $n=30$; multiplicities change.

8 Cross-analysis vs H4 / H5

H4's ~ 5 k-line / ~ 70 s root no-sols are the $M=10$ endpoint of H6a (ten failed-band replays). Secondary H5 pointer: **greedy_cautious** H0 a reaches the same no-sol in ≈ 344 lines / 0.42s via a different search bundle — H4 cost is not “cautious alone.”

9 Boundaries / non-claims

- Not a Bucket 3 claim.
- Not “cautious is better.”
- Root no-solution is not mechanism recovery.
- Budget-relative no-solution $\neq$ unlearnability at larger M .
- Control c solved / delta match is not automatic causal recovery.
- No $n \times \text{folding_steps}$ interaction claim (grid not run).
- Approximate linearity only in these limited slices.
- H7a and H7b analysed separately; H0–H5 readings preserved.

10 Next

H6 documentation complete. H7a analysed (`h7a_relto_vs_sechk.md`); H7b analysed separately (`h7b_cautious_relto_vs_sechk.md`). Still **no Bucket 3 claim**.